SHIVAM OJHA

New York, NY | shivam.ojha11@gmail.com | LinkedIn: @shivam-ojha | GitHub: @shivamojha2 | +1 (917) 302-7544

SUMMARY

Research Engineer focused on LLM post-training, inference optimization, and scalable agentic AI systems.

TECHNICAL SKILLS

- LLM Research & Post-Training: GRPO, RLVR, QLoRA, Fine-Tuning, Automated Verification, Trajectory Evaluation
- Inference Optimization: Continuous Batching, Prefix Caching, Quantization, Speculative Decoding, MoE Pruning
- Agentic AI Systems: Multi-Agent Orchestration, Dynamic Routing, Tool Execution & Error Recovery, Agent Memory, Context Management, RAG
- Frameworks & Infrastructure: PyTorch, Hugging Face Transformers, vLLM, PyTorch FSDP, Ray Serve, LangGraph, LangChain, Docker, Kubernetes, AWS (SageMaker, Bedrock, Textract), Elasticsearch, Airflow

WORK EXPERIENCE

Principal Research Engineer, LLM Core & Agentic AI | Capital One (New York, NY) Jan 2026 – Present

- Developing inference optimizations for agentic workloads through GPT-OSS expert pruning and MoE expert-utilization analysis; benchmarking and tuning vLLM serving configurations, including continuous batching, prefix caching, speculative decoding, quantization, and KV-cache tuning, using TTFT, TPOT, throughput, and end-to-end agent latency to characterize performance trade-offs.
- Built a configurable multi-agent orchestration SDK supporting sequential execution and dynamic routing, with agent state and context management, enabling downstream teams to develop and productionize agentic applications, by composing reusable skills and tools.
- Designed a pluggable agent memory system informed by FluxMem and A-MEM, supporting adaptive memory organization, retrieval, and updates based on interaction patterns.
- Implemented modular self-reflection loops inspired by ACE's generator–reflector–curator architecture, evaluating execution trajectories and transforming feedback into distilled lessons consolidated in an evolving context playbook to support agent adaptation.
- Designed and implemented reward functions for agentic reinforcement learning, exploring reward signals for agent behavior and skill acquisition informed by SkillRL.

Senior Machine Learning Engineer, R&D | CCC Intelligent Solutions (Chicago, IL) Jun 2022 – Dec 2025

- Trained policy-constrained LLMs using RLVR with GRPO, optimizing group-sampled reasoning trajectories against automated insurance-policy, consistency, and grounding verifiers to improve compliance and reasoning reliability.
- Led development of in-house LLM training and serving infrastructure spanning dataset curation, QLoRA fine-tuning, evaluation harnesses, and multi-node deployments; optimized serving throughput and latency with vLLM for Llama 4 and Qwen models.
- Developed a repair assistant agent using agentic RAG, automated retrieval evaluation, safety guardrails, and a voice interface. Integrated Whisper for speech-to-text, citation-based retrieval, and ReAct-style reasoning to deliver grounded, auditable responses.
- Designed and trained ViT-based computer vision models to estimate chrome panel damage and other part-level vehicle attributes, incorporating custom loss functions to disambiguate visually similar operations

Data Science & Cognitive Analyst | Deloitte (Bengaluru, India) Aug 2019 - Jul 2021

- Conducted rigorous EDA on medical datasets to identify key predictive features for patient outcomes, establishing performance baselines with POC models in AWS SageMaker to guide future development.

EDUCATION

Columbia University New York, NY
M.S. in Electrical Engineering, Specialization in Machine Learning, CGPA: 4.09/4 Dec 2022
Awards - MS Honors Student Award; MS EE Award of Excellence

Manipal Academy of Higher Education, Manipal Institute of Technology Manipal, India
B.Tech. in Electrical and Electronics Engineering, CGPA: 8.35/10 May 2019

PUBLICATIONS

- **Real-time Multi-Camera Analytics for Traffic Information Extraction and Visualization** | 2023 IEEE International Conference on Pervasive Computing, March 2023